



Unit I: Probability

Definitions of Probability, Set Operations,
Conditional Probability, Independence & Bayes' Theorem

STA201 – Theory of Probability and Statistics

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Dr. Lokesh Singh

Department of Mathematics,
Amity University Punjab, Mohali.

lsingh@pb.amity.edu

1 Introduction

Probability theory provides a mathematical framework to quantify uncertainty. For engineers, it is the language used to model:

- Reliability of systems (e.g., probability a component fails within a certain time),
- Quality control (e.g., probability a batch contains more than 2 defective items),
- Signal processing (e.g., probability of bit error in a communication channel),
- Risk analysis (e.g., probability of structural failure under extreme loads).

In this unit we build the set-theoretic foundations of probability and study the rules that govern the calculus of chance. Every example is chosen so that later we can attach a **random variable** (a numerical summary) to the outcomes, enabling us to compute expectations, variances, and full probability distributions (Unit II).

2 Random Experiments

A **random experiment** is a process that leads to one of several possible outcomes, where the outcome cannot be predicted with certainty beforehand. Examples: tossing a coin, rolling a die, measuring the lifetime of a battery, counting the number of defective items in a sample.

3 Sample Space

The **sample space** of a random experiment, denoted by S , is the set of all possible outcomes.

Examples:

- Tossing a coin: $S = \{H, T\}$.
- Tossing two coins: $S = \{HH, HT, TH, TT\}$.
- Rolling a die: $S = \{1, 2, 3, 4, 5, 6\}$.
- Lifetime of a component (in hours): $S = [0, \infty)$ (continuous).

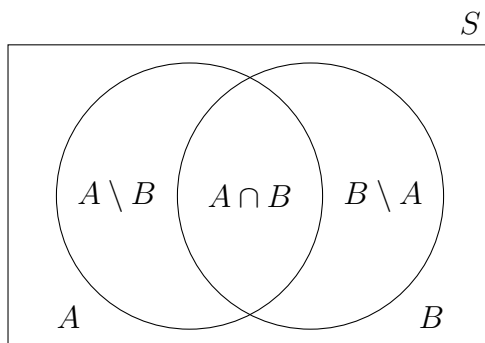
4 Events

An **event** is any subset of the sample space S . We say the event *occurs* if the actual outcome of the experiment belongs to that subset.

Set Operations and Venn Diagrams

For events A and B in S :

- **Union** $A \cup B$: outcomes in A or B or both.
- **Intersection** $A \cap B$: outcomes in both A and B .
- **Complement** $A' = \bar{A} = S \setminus A$: outcomes not in A .
- Two events are **mutually exclusive** (disjoint) if $A \cap B = \emptyset$ (they cannot happen together).

Figure 1: Venn diagram of events A and B in a sample space S .

5 Definitions of Probability

5.1 Classical Definition (Equally Likely Outcomes)

If the sample space S consists of a finite number N of outcomes that are **equally likely**, then for any event $A \subseteq S$,

$$P(A) = \frac{|A|}{|S|} = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}}.$$

This definition assumes symmetry – fair coins, fair dice, well-shuffled cards, random selection from a batch.

5.2 Axiomatic Definition (Kolmogorov, 1933)

A probability measure P is a function that assigns to each event $A \subseteq S$ a real number such that:

1. **Non-negativity:** $P(A) \geq 0$ for every event A .
2. **Unit measure:** $P(S) = 1$.
3. **Countable additivity:** If $A_1, A_2, \dots$ are mutually exclusive events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

From these axioms we derive:

- $P(\emptyset) = 0$.
- $P(A') = 1 - P(A)$.
- Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- If $A \subseteq B$, then $P(A) \leq P(B)$.

6 Set-Theoretic Probability Examples

Solved Example Tossing Two Coins

Two fair coins are tossed. Sample space:

$$S = \{HH, HT, TH, TT\}, \quad |S| = 4.$$

Define events:

- A : both coins show the same face. $A = \{HH, TT\}$.
- B : at least one head occurs. $B = \{HH, HT, TH\}$.
- C : exactly one head occurs. $C = \{HT, TH\}$.

Find:

- (a) $P(A)$, $P(B)$, $P(C)$
- (b) $P(A \cap B)$ (same face and at least one head)
- (c) $P(A \cup B)$
- (d) $P(A' \cap B)$ (not same face and at least one head)

Solution:

- (a) $P(A) = 2/4 = 1/2$, $P(B) = 3/4$, $P(C) = 2/4 = 1/2$.
- (b) $A \cap B = \{HH\}$, $P(A \cap B) = 1/4$.
- (c) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{2} + \frac{3}{4} - \frac{1}{4} = 1$ (indeed $A \cup B = S$).
- (d) $A' = \{HT, TH\}$, $A' \cap B = \{HT, TH\} = C$, $P = 1/2$.

Solved Example Rolling a Fair Die

A fair six-sided die is rolled. $S = \{1, 2, 3, 4, 5, 6\}$, $|S| = 6$. Define:

- A : even number, $A = \{2, 4, 6\}$.
- B : number > 3 , $B = \{4, 5, 6\}$.
- C : prime number, $C = \{2, 3, 5\}$.

Compute:

- (a) $P(A)$, $P(B)$, $P(C)$
- (b) $P(A \cap B)$ (even and > 3)
- (c) $P(A \cup B)$
- (d) $P(A' \cap C)$ (odd and prime)

Solution:

- (a) $P(A) = 3/6 = 1/2$, $P(B) = 3/6 = 1/2$, $P(C) = 3/6 = 1/2$.
- (b) $A \cap B = \{4, 6\}$, $P = 2/6 = 1/3$.
- (c) $A \cup B = \{2, 4, 5, 6\}$, $P = 4/6 = 2/3$. (Addition: $\frac{1}{2} + \frac{1}{2} - \frac{1}{3} = 1 - \frac{1}{3} = 2/3$.)
- (d) $A' = \{1, 3, 5\}$, $A' \cap C = \{3, 5\}$, $P = 2/6 = 1/3$.

Solved Example Drawing Cards

One card from a well-shuffled standard 52-card deck. Define:

- H : Heart (13 cards),
- K : King (4 cards),
- F : face card (J, Q, K; 12 cards).

Find:

- $P(H), P(K), P(F)$
- $P(H \cap K)$ (King of Hearts)
- $P(H \cup K)$
- $P(F' \cap H)$ (Heart, not face)

Solution:

- $P(H) = 13/52 = 1/4, P(K) = 4/52 = 1/13, P(F) = 12/52 = 3/13.$
- $H \cap K$: King of Hearts $\Rightarrow 1/52.$
- $P(H \cup K) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}.$
- F' : non-face (40 cards). Heart non-face: Ace–10 of Hearts (10 cards). $P = 10/52 = 5/26.$

Solved Example Manufacturing Defects

1000 components: 50 have defect A, 80 have defect B, 20 have both. Random selection.

- A : component has defect A,
- B : component has defect B.

Compute:

- $P(A), P(B), P(A \cap B)$
- $P(A \cup B)$ (at least one defect)
- $P(A' \cap B')$ (neither defect)

Solution: $P(A) = 50/1000 = 0.05, P(B) = 0.08, P(A \cap B) = 0.02.$

- see above.
- $P(A \cup B) = 0.05 + 0.08 - 0.02 = 0.11.$
- $P(A' \cap B') = P((A \cup B)') = 1 - 0.11 = 0.89.$

Solved Example Two Dice

Two fair dice (red and blue). $S = \{(i, j) \mid 1 \leq i, j \leq 6\}, |S| = 36.$ Define:

- E : sum is even,

- T : sum ≥ 8 ,
- D : doubles (both numbers equal).

Find:

- $P(E), P(T), P(D)$
- $P(E \cap D)$ (even sum and doubles)
- $P(T \cup D)$
- $P(E' \cap T)$ (odd sum and sum ≥ 8)

Solution (sketch):

- Even sums: 2,4,6,8,10,12 $\rightarrow$ 18 outcomes, $P(E) = 1/2$; sum ≥ 8 : 15 outcomes, $P(T) = 5/12$; doubles: 6 outcomes, $P(D) = 1/6$.
- $E \cap D$: (2,2),(4,4),(6,6) $\rightarrow$ 3 outcomes, $P = 3/36 = 1/12$.
- $T \cap D$: (4,4),(5,5),(6,6) $\rightarrow$ 3 outcomes, so $P(T \cup D) = \frac{15}{36} + \frac{6}{36} - \frac{3}{36} = \frac{18}{36} = 1/2$.
- Odd sums ≥ 8 : sums 9 (4 outcomes) and 11 (2 outcomes) $\rightarrow$ 6 outcomes, $P = 6/36 = 1/6$.

(Full count can be elaborated in class.)

7 Conditional Probability

If we know that event B has occurred, the sample space is reduced to B . The **conditional probability** of A given B is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

This leads to the **multiplication rule**:

$$P(A \cap B) = P(A | B) P(B) = P(B | A) P(A).$$

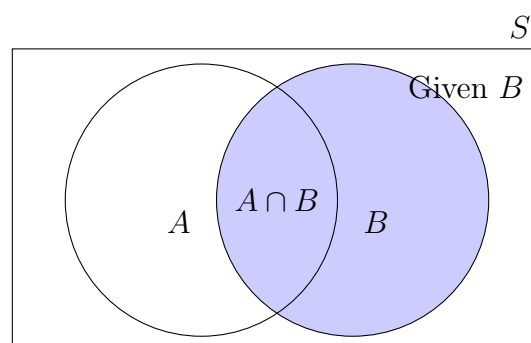


Figure 2: Conditional probability reduces the sample space to B .

Solved Examples on Conditional Probability

Solved Example Defects (revisited)

Using the data from Example 4 (1000 components: $P(A) = 0.05$, $P(B) = 0.08$, $P(A \cap B) = 0.02$), find the probability that a component has defect A given it has defect B.

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.02}{0.08} = 0.25.$$

Likewise, $P(B | A) = \frac{0.02}{0.05} = 0.40$. So among A-defective parts, 40% also have B.

Solved Example Fuses

A lot of 100 fuses contains 10 defectives. Two fuses are drawn without replacement.

- (a) What is the probability that the second fuse is defective given the first is defective?
- (b) What is the probability that both are defective?

Solution: Let D_1 : first defective, D_2 : second defective.

- (a) After drawing a defective first, 9 defectives remain out of 99, so $P(D_2 | D_1) = 9/99 = 1/11$.
- (b) $P(D_1 \cap D_2) = P(D_1)P(D_2 | D_1) = \frac{10}{100} \cdot \frac{9}{99} = \frac{1}{110}$.

Solved Example Cards and Conditional

From a standard deck, one card is drawn. Let Q be the event “Queen” and H “Heart”. Find $P(Q | H)$ and $P(H | Q)$.

$$P(Q) = 4/52 = 1/13, P(H) = 13/52 = 1/4, P(Q \cap H) = 1/52.$$

$$P(Q | H) = \frac{1/52}{13/52} = \frac{1}{13} = P(Q) \quad (\text{independence discussed later}),$$

$$P(H | Q) = \frac{1/52}{4/52} = \frac{1}{4} = P(H).$$

Solved Example Weather and Failure

A sensor fails with probability 0.1 under normal weather and 0.25 under extreme weather. The probability of extreme weather on any day is 0.05. If the sensor fails today, what is the probability that the weather is extreme?

$$P(\text{extreme} | \text{fail}) = \frac{P(\text{fail} | \text{extreme})P(\text{extreme})}{P(\text{fail})}.$$

First compute total failure probability:

$$P(\text{fail}) = 0.25 \cdot 0.05 + 0.1 \cdot 0.95 = 0.0125 + 0.095 = 0.1075.$$

Then

$$P(\text{extreme} \mid \text{fail}) = \frac{0.0125}{0.1075} \approx 0.1163.$$

(This also previews Bayes' theorem.)

8 Independence

Two events A and B are **independent** if the occurrence of one does not affect the probability of the other:

$$P(A \cap B) = P(A)P(B).$$

Equivalently, $P(A \mid B) = P(A)$ (when $P(B) > 0$).

Note: Independence is not the same as mutual exclusivity. Mutually exclusive events with non-zero probability are always *dependent* because $P(A \mid B) = 0 \neq P(A)$.

Solved Examples on Independence

Solved Example Series System

A circuit has two independent components in series. The system works only if both work. If $p_1 = 0.9$ and $p_2 = 0.85$ are the probabilities each component works, the reliability is

$$P(\text{system works}) = P(\text{comp1 works} \cap \text{comp2 works}) = 0.9 \times 0.85 = 0.765.$$

Solved Example Parallel System

The same components in parallel: system works if at least one works.

$$P(\text{system works}) = 1 - P(\text{both fail}) = 1 - (1 - 0.9)(1 - 0.85) = 1 - 0.1 \times 0.15 = 0.985.$$

Solved Example Cards Independence Check

From a deck, events H (Heart) and K (King) are independent because

$$P(H \cap K) = 1/52, \quad P(H)P(K) = (13/52) \cdot (4/52) = 52/2704 = 1/52.$$

Solved Example Dice Independence

Roll two fair dice. Let A : first die shows 6, B : sum is odd. Determine if A and B are independent.

$$P(A) = 1/6, \quad P(B) = 18/36 = 1/2.$$

$A \cap B$: first die 6 and sum odd $\rightarrow$ second die must be odd (1,3,5) $\rightarrow$ 3 outcomes, $P(A \cap B) = 3/36 = 1/12$. Check: $P(A)P(B) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12} = P(A \cap B)$, so **independent**.

9 Bayes' Theorem

9.1 Introduction

Bayes' theorem is a fundamental result in probability theory that allows us to **update** the probability of an event based on new evidence. It is the backbone of many engineering applications such as diagnostic testing, spam filtering, machine learning classification, reliability analysis, and fault detection.

9.2 The Formula

For events A and B with $P(B) > 0$:

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}.$$

More generally, if $A_1, A_2, \dots, A_k$ form a partition of the sample space S (mutually exclusive and exhaustive) and B is any event with $P(B) > 0$, then for any i :

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{\sum_{j=1}^k P(B | A_j) P(A_j)}.$$

9.3 Interpretation of Terms

- **Prior probability** $P(A_i)$ – initial degree of belief about A_i before observing evidence B .
- **Likelihood** $P(B | A_i)$ – probability of observing evidence B assuming A_i is true.
- **Evidence / Marginal** $P(B) = \sum_j P(B | A_j) P(A_j)$ – total probability of observing B under all possible causes.
- **Posterior probability** $P(A_i | B)$ – updated belief about A_i after observing B .

Bayes' theorem essentially says:

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}.$$

9.4 Derivation

Recall the definition of conditional probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad \text{and} \quad P(B | A) = \frac{P(A \cap B)}{P(A)}.$$

Multiplying the second equation by $P(A)$ gives $P(A \cap B) = P(B | A)P(A)$. Substituting this into the first equation:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

Thus Bayes' theorem is a direct consequence of the conditional probability definition.

9.5 Law of Total Probability

If $A_1, A_2, \dots, A_k$ are mutually exclusive and exhaustive events, then

$$P(B) = \sum_{j=1}^k P(B \cap A_j) = \sum_{j=1}^k P(B | A_j)P(A_j).$$

This forms the denominator in Bayes' theorem.

9.6 Step-by-Step Procedure

1. Identify the partition $A_1, A_2, \dots, A_k$ (possible causes/hypotheses).
2. Write down the prior probabilities $P(A_j)$.
3. Identify the evidence B .
4. Write down the likelihoods $P(B | A_j)$.
5. Compute the total probability $P(B)$ using the law of total probability.
6. Apply Bayes' formula to find the posterior $P(A_i | B)$.
7. Interpret the result.

9.7 Solved Examples

Solved Example Defective Parts from Two Suppliers

A manufacturer buys components from two suppliers. Supplier X provides 70% of parts with a defect rate of 2%; Supplier Y provides 30% with a defect rate of 5%. A part is selected at random and found defective. What is the probability it came from Supplier Y?

Solution: Let X = part from X, Y = part from Y, D = defective.

$$P(X) = 0.7, P(Y) = 0.3, P(D|X) = 0.02, P(D|Y) = 0.05.$$

Total probability:

$$P(D) = P(D|X)P(X) + P(D|Y)P(Y) = 0.02 \cdot 0.7 + 0.05 \cdot 0.3 = 0.014 + 0.015 = 0.029.$$

Bayes:

$$P(Y|D) = \frac{P(D|Y)P(Y)}{P(D)} = \frac{0.05 \cdot 0.3}{0.029} = \frac{0.015}{0.029} \approx 0.517.$$

Even though Y supplies only 30%, it accounts for over half the defects due to its higher defect rate.

Solved Example Medical Diagnosis with a Rare Disease

A diagnostic test for a disease has sensitivity $P(+|D) = 0.95$ and specificity $P(-|D') = 0.90$. The prevalence is $P(D) = 0.02$. If a person tests positive, what is the probability they actually have the disease?

Solution:

$$P(+) = 0.95 \cdot 0.02 + 0.10 \cdot 0.98 = 0.019 + 0.098 = 0.117.$$

$$P(D|+) = \frac{0.95 \cdot 0.02}{0.117} = \frac{0.019}{0.117} \approx 0.162.$$

Despite an apparently accurate test, only about 16% of positive results correspond to actual disease. This is the base rate fallacy.

Solved Example Email Spam Filter

Suppose 20% of all emails are spam. 95% of spam emails contain the word “offer”, while only 5% of legitimate emails contain this word. If an email contains “offer”, what is the probability that it is spam?

Solution: Let S = spam, O = word “offer”.

$$P(S) = 0.2, P(S') = 0.8, P(O|S) = 0.95, P(O|S') = 0.05.$$

$$P(O) = 0.95 \cdot 0.2 + 0.05 \cdot 0.8 = 0.19 + 0.04 = 0.23.$$

$$P(S|O) = \frac{0.95 \cdot 0.2}{0.23} = \frac{0.19}{0.23} \approx 0.826.$$

The word “offer” raises the probability of spam from 20% to about 83%.

Solved Example Paint-Store Chain (from Sample Questions)

A paint-store chain sells latex and semigloss paint. The probability that a customer purchases latex paint is 0.75. Of those who purchase latex paint, 60% also purchase rollers; only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Solution: Let L = latex paint, L' = semigloss, R = roller.

$$P(L) = 0.75, P(L') = 0.25, P(R|L) = 0.60, P(R|L') = 0.30.$$

Total probability of roller:

$$P(R) = 0.75 \cdot 0.60 + 0.25 \cdot 0.30 = 0.45 + 0.075 = 0.525.$$

Bayes:

$$P(L|R) = \frac{P(R|L)P(L)}{P(R)} = \frac{0.45}{0.525} = \frac{6}{7} \approx 0.857.$$

Thus, given a roller purchase, there is an 85.7% chance the paint is latex.

Solved Example Manufacturing Plans (from Sample Questions)

A firm uses three plans P_1, P_2, P_3 for 30%, 20%, and 50% of products. The defect rates are $P(D|P_1) = 0.01$, $P(D|P_2) = 0.03$, $P(D|P_3) = 0.02$. A randomly selected product is defective. Which plan was most likely used?

Solution: Priors: $P(P_1) = 0.30$, $P(P_2) = 0.20$, $P(P_3) = 0.50$. Total defective probability:

$$P(D) = 0.30 \cdot 0.01 + 0.20 \cdot 0.03 + 0.50 \cdot 0.02 = 0.003 + 0.006 + 0.010 = 0.019.$$

Posterior probabilities:

$$P(P_1|D) = \frac{0.003}{0.019} \approx 0.158, \quad P(P_2|D) = \frac{0.006}{0.019} \approx 0.316, \quad P(P_3|D) = \frac{0.010}{0.019} \approx 0.526.$$

Plan 3 is most likely to have produced the defective item.

Solved Example Assembly Plant Machines (Law of Total Probability)

In an assembly plant, machines B_1, B_2 , and B_3 produce 30%, 45%, and 25% of products. Their defective rates are 2%, 3%, and 2%. A finished product is randomly selected. What is the probability that it is defective?

Solution:

$$P(D) = 0.30 \cdot 0.02 + 0.45 \cdot 0.03 + 0.25 \cdot 0.02 = 0.006 + 0.0135 + 0.005 = 0.0245.$$

This is an application of the law of total probability, which is the denominator in Bayes' theorem.

Solved Example Structural Flaw Detection

A non-destructive test for a structural flaw has sensitivity 95% and specificity 90%. The prior probability of a flaw is 2%. Find $P(\text{flaw} | \text{positive})$.

$$\begin{aligned} P(F) &= 0.02, \quad P(+ | F) = 0.95, \\ P(+ | \bar{F}) &= 0.10. \end{aligned}$$

Total positive probability:

$$P(+) = 0.02 \times 0.95 + 0.98 \times 0.10 = 0.019 + 0.098 = 0.117.$$

$$P(F | +) = \frac{0.019}{0.117} \approx 0.1624.$$

Again, a positive result is far from certain.

9.8 Engineering Importance

Bayes' theorem is used in reliability engineering, signal processing (Kalman filters), machine learning (Naive Bayes classifiers), and decision theory.

10 From Events to Random Variables (Preview)

Many of our examples count occurrences or flag conditions. In Unit II, we will define a **random variable** X mapping outcomes to numbers:

- Two-coin toss: X = number of heads; $P(X = 1) = P(\{HT, TH\}) = 1/2$.
- Die: Y = face value; $P(Y \text{ even}) = P(\{2, 4, 6\})$.
- Defect inspection: $Z = 1$ if defect A present, else 0; a Bernoulli random variable with $p = 0.05$.

All probability rules (union, intersection, complement, conditional, independence) will be applied directly to events expressed in terms of random variables.

Key Formulas Summary

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A') = 1 - P(A)$
- $P(A | B) = \frac{P(A \cap B)}{P(B)}$
- $P(A \cap B) = P(A | B)P(B)$
- Independence: $P(A \cap B) = P(A)P(B)$
- Total probability: $P(A) = \sum_i P(A | B_i)P(B_i)$
- Bayes: $P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_j P(A | B_j)P(B_j)}$

Practice Problems

1. Three fair coins are tossed. Write the sample space. Let A : at least two heads; B : the first coin is head. Compute $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, and $P(B | A)$.
2. A card is drawn from a well-shuffled deck. Let A be “face card” and B be “Spade”. Compute $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, and verify if A and B are independent.

3. In a group of 200 students, 120 study Mathematics, 90 study Physics, and 50 study both. Find the probability that a randomly selected student studies (i) neither, (ii) only Mathematics, (iii) only one subject.
4. Two dice are rolled. Find the probability that the number on the red die is strictly greater than that on the blue die.
5. A circuit has two independent switches in series. Each switch has probability 0.02 of being open. What is the probability the circuit allows current?
6. In the supplier defect example, compute $P(X | D)$ and interpret.
7. A diagnostic test for a software bug has sensitivity 98% and specificity 96%. If 1% of modules contain a bug, what is the probability a module flagged by the test actually has a bug?
8. A manufacturer uses three machines that produce 40%, 35%, and 25% of parts, with defect rates 1%, 2%, and 3%, respectively. A randomly selected part is defective; what is the probability it came from the third machine?